

Spiking Reset for Improved Accuracy and Robustness to Timing Perturbation

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Parallel spiking neurons drop the membrane reset, computing timesteps simultaneously. At identical parameter count, removal costs forty accuracy points and 3.3x the synaptic operations; at matched accuracy the reset appears to add robustness to timing and order perturbation but does not appear to add robustness to spike insertion or deletion.

I. Introduction

A leaky integrate-and-fire neuron decays its membrane potential, accumulates input into it, and, when the potential crosses a threshold, emits a spike and resets the potential. The reset is a conditional and state-dependent subtraction; it is the only nonlinearity in the recurrent path. That component is now being removed in practice. With the reset, each step's update depends on whether the membrane crossed the threshold at the step before, a condition on the state itself, so the steps have to be computed one after another. A neuron whose membrane is never reset has a linear state update, and a linear recurrence can be evaluated for every step at once by a parallel scan, which is the explicit motivation for parallel spiking neurons [1] and the work following them. Some work also reports spiking networks as inherently robust to input perturbation [2], without settling the mechanism. So one line of work removes the reset for speed while another looks for the source of robustness without isolating it, and neither what the reset contributes to robustness nor what removing it costs in accuracy appears to have been settled.

Two earlier ablation studies come closest to the question. “Towards Understanding the Effect of Leak in Spiking Neural Networks” [3] isolates the membrane leak and reads a mechanism, low-pass filtering, rather than a benchmark number; it is the template this work follows, and it holds the reset fixed throughout. “Understanding the functional roles of modelling components in spiking neural networks” [4] ablates leakage, reset and recurrence together and reports the reset as having little effect on robustness, from single numbers on one dataset, without seeds and without matching the compared networks on clean accuracy, their accuracy on unperturbed input. The spike-perturbation operators used here are taken unchanged from “Are spiking neural networks exploiting temporal information on neuromorphic benchmarks?” [5], which should keep the measurement comparable rather than bespoke.

This work ablates the reset alone, on two axes kept apart: what it does for clean accuracy at fixed capacity, and what it still does for robustness once clean accuracy is matched. The reset is a dynamical rule and not a weight, so the networks compared here have identical parameter counts, identical initialisations and identical training budgets, and every network is retrained from scratch rather than mutated at test time.

II. Methods and Results

Three variants of one leaky integrate-and-fire network are trained, differing only in what happens to the membrane after a spike: the soft-reset variant subtracts the threshold, the hard-reset variant sets the membrane to zero, and the reset-removed variant leaves it unchanged. Spike-encoded modular counting (chance 20%) sits on an expressivity boundary, which gives the accuracy axis a prediction before any measurement: a non-leaky integrate-and-fire unit with threshold t emits $\text{floor}(N/t)$ spikes after N input events, so a linear readout of a population of such units recovers $N \bmod t$ at any sequence length. The reset makes the neuron a counter. Remove it and the same unit is a nonnegative-diagonal linear recurrence, which can count without bound but, on this argument, cannot take a residue. The naturalistic tier is the Spiking Heidelberg Digits (SHD) and Spiking Speech Commands (SSC) [6]. The networks are shallow feedforward stacks of leaky integrate-and-fire units trained with surrogate gradients, up to five seeds per variant.

After clean training, every variant is evaluated on the same perturbed copies of the test set, so within a point the variants differ only in the reset. Six operators cover three classes: count-changing (insert, delete), order-changing (transpose, time reversal) and timing (jitter, dilate) [5]. Perturbation is read at a matched clean-accuracy operating point, because a network closer to chance has less fitted structure to disturb and so tends to read as less sensitive to any perturbation: training stores snapshots as accuracy rises, and every variant is re-swept from the snapshot nearest the target accuracy.

The reset appears to have a large effect on accuracy, in the predicted direction (Fig. 1a). On modular counting the soft-reset variant reaches 100% while the reset-removed variant's best is 36%, a few points above the 30% ceiling of a memoriser that recognises the test rasters occurring verbatim in training and guesses on the rest: it does not appear to learn to count within the training budget. On SHD, removing the reset costs 42 points of clean accuracy (78% against 36%), and on SSC it costs almost the whole task (37% and 30% for the soft and hard variants against 5.7% at a 2.9% chance floor). Because the variants are parameter-identical, none of this gap is parameter count.

Removing the reset also costs activation sparsity, and the cost appears to grow with sequence length, since a unit that is never reset fires at every step once it is past the threshold: at identical dense FLOPs the reset-removed variant draws 1.2 times the soft variant's synaptic operations on the short counting task and 3.3 times on the longer SHD recordings, so the parallel-scan speed-up comes with several times the spike traffic, the usual proxy for energy on neuromorphic hardware.

Matched on clean accuracy, that is, with the variants compared at equal accuracy on unperturbed input, the reset still appears to add certain forms of robustness (Fig. 1b). Swept at the reset-removed variant's best accuracy on SHD, where all three variants sit between 36% and 43% clean, the soft and hard variants retain 81% to 97% of that accuracy under timing jitter and dilation against 23% to 38% for the reset-removed variant; under complete time reversal they hold 71% and 60% against 13%. Against the count-changing operators no such advantage appears: insertion, at the budget shown, costs every variant most of its accuracy (12%, 11% and 26% retained, all three near the chance floor). The split is consistent with the counting mechanism: a reset-bearing unit's state is a spike count, which is invariant to rearrangement in time and moved only by adding or removing spikes.

III. Conclusions

Removing the reset so that the membrane update can be computed in parallel does not appear to be free: at identical parameter count it costs most of the task on all three datasets and up to several times the spike traffic. On the evidence so far, the reset also appears to contribute robustness beyond the accuracy it brings, but of specific kinds, so a robustness claim about spiking networks should name the perturbation class it holds for and be read at a matched operating point.

IV. References

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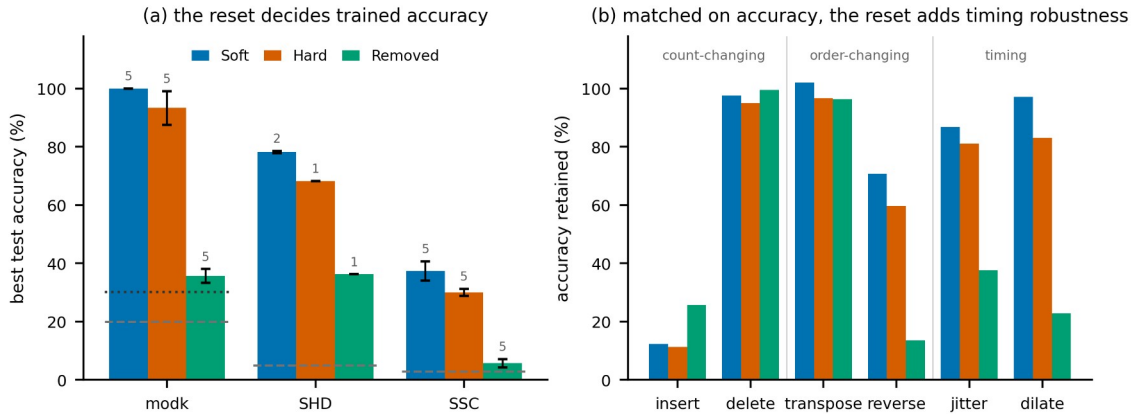


Figure 1. The spiking reset at identical parameter count. (a) Best test accuracy per variant; seeds above each bar, dashed lines chance, dotted line the 30% memoriser floor. (b) The same variants on SHD, matched at the reset-removed variant's best clean accuracy (soft 39%, hard 43%, removed 36%): the accuracy each retains per operator.